

Data Inventory and Quality Assessment

Purpose

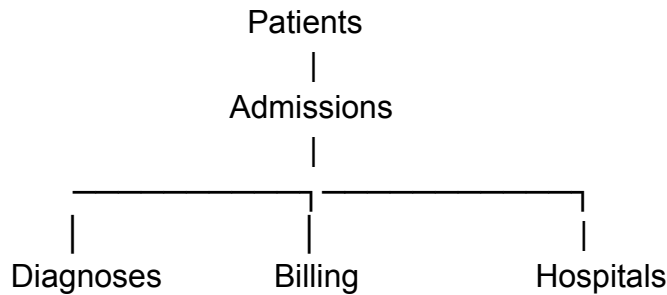
Establish an inventory of the source datasets, document their structure and relationships, assess data quality, and identify material issues that could affect downstream analysis, modeling, and reporting. The assessment also establishes how identified issues will be addressed during ETL.

Source Data Inventory and Quality Summary

Assessment	Datasets				
	patients	admissions	diagnoses	billing	hospitals
Description	Patient demographic master table	Hospital admission records	Disnosis records linked to admission	Financial billing linked to admission	Hospital facility reference table
Primary Key	patient_id	Admission_id,	diag_id	bill_id	hospital_id
Foreign Key	N/A	Patient_id, hospital_id	admission_id	admission_id	N/A
Total Rows	86400	120000	271341	120000	33
Total Columns	8	17	6	6	6
Missing Values	24715	0	0	0	0
Missing(%)	3.58	0	0	0	0
Duplicate Rows	0	0	0	0	0
Invalid Values	0	0	0	0	0
Primary Key Unique	pass	pass	pass	pass	pass

Assessment	Datasets				
	patients	admissions	diagnoses	billing	hospitals
Referential Integrity	pass	pass	pass	pass	pass
Overall Quality	Good	Good	Excellent	Excellent	Excellent

Entity Relationship Diagram (ERD)



The admissions table serves as the central transactional table, linking patient, hospital, diagnosis, and billing information. Diagnoses is a one-to-many relationship with admissions, while patients, hospitals, and billing provide supporting one-to-one or many-to-one relationships as defined by the source data.

Key Findings

- Overall data quality is **Good**, with no duplicate records, primary key issues, or referential integrity violations identified.
- patients.insurance_type contains **24,715 missing values (3.58%)**; no other material missing-data issues were identified.
- The diagnoses table contains multiple diagnosis records for some admissions; **49 admissions have four diagnosis records each**. Diagnosis-level analyses will account

for these multiple diagnoses by evaluating unique admission–ICD-10 code combinations.

- The hospitals dataset contains **33 unique hospital_id values but only 26 unique hospital names**, indicating that hospital names alone do not uniquely identify all hospitals.
- Some hospitals share the same name but have different bed counts. A derived hospital_name field combining **hospital name + bed count** was created to provide a distinguishable hospital label for downstream analysis and dashboard reporting.
- admissions.admit_date and discharge_date are stored as strings and require datatype conversion.
- **Readmission eligibility:** Admissions ending in an expired discharge were excluded from readmission-rate calculations because these patients were not eligible for a subsequent post-discharge readmission. A total of 7,413 admissions (6.18%) had an expired discharge status. For descriptive reporting, readmission rates were therefore calculated using eligible discharges as the denominator. The predictive modeling cohort similarly excludes expired admissions.
- The identified issues are **minor and manageable** and are not expected to prevent downstream analysis.

Limitations:

- 30-day mortality after discharge could not be assessed because the dataset does not contain post-discharge mortality information. Consequently, patients coded as not readmitted may include individuals who died within 30 days without being readmitted.

Recommendations

- **Proceed with the dataset** for the healthcare readmission analytics engagement.
- Retain patient records with missing insurance_type; exclude missing values only from analyses specifically requiring insurance information.
- Convert admit_date and discharge_date to appropriate date/datetime types during BigQuery SQL ETL.
- For diagnosis-level analyses, treat each unique **admission–ICD-10 code combination** as a separate observation to preserve multiple diagnoses associated with the same admission without duplicating diagnosis records.
- Use **hospital_id as the primary hospital identifier** for joins and analytical calculations rather than relying on hospital names alone.

- Use the derived **hospital_name (name + bed count)** as a human-readable label for hospital-level reporting and Tableau visualizations.
 - Preserve the raw tables unchanged and perform transformations in staging/analytical tables.
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Overall Assessment

Data is suitable for analysis following minor preprocessing.

Detailed Column Dictionary

Dataset: patients

Column name	Data Type	Expected Data Type	Missing %	Invalid Value	Description
patient_id	string	string	0	0	Unique anonymised patient identifier. Reused across multiple admissions.
age	integer	integer	0	0	Age at admission (years)
gender	string	string	0	0	Gender: M / F / Other
state	string	string	0	0	Indian state of patient residence. 15 states covering ~88% of India's population.
bpl_card	boolean	boolean	0	0	Whether the patient holds a Below Poverty Line ration card (True/False)
insurance_type	string	string	28.6	24715 null	Health insurance coverage
comorbidity_count	integer	integer	0	0	Number of chronic conditions.
prev_admissions	integer	integer	0	0	Number of hospital admissions in the prior 12 months.

Dataset: admissions

Column name	Data Type	Expected Data Type	Missing %	Invalid Value	Description
admission_id	string	string	0	0	Primary key. Unique identifier for each hospital admission.
patient_id	string	string	0	0	Foreign key → patients.csv. Links to the patient's demographic record.
admit_date	string	datetime	0	0	Date of admission (YYYY-MM-DD). Ranges 2015-01-01 to 2024-12-31.
discharge_date	string	datetime	0	0	Date of discharge. Always ≥ admit_date.
los_days	integer	integer	0	0	Length of stay in days (derived: discharge_date – admit_date). Range: 1–90.
admit_type	string	string	0	0	How the patient was admitted: Emergency / Elective / OPD.
ward_type	string	string	0	0	Ward where patient was treated: General / ICU / HDU / NICU.
hospital_id	string	string	0	0	Foreign key → hospitals.csv. Identifies the treating hospital.
discharge_type	string	string	0	0	Discharge outcome: Recovered / LAMA (Left Against Medical Advice) / Expired / Referred.
num_procedures	integer	integer	0	0	Number of clinical procedures performed during admission. Range: 0–8. Correlated with ward type and comorbidities.
charlson_index	integer	integer	0	0	Charlson Comorbidity Index score computed from ICD-10 diagnoses and patient age. Higher = greater mortality risk. Range: 0–12.
hba1c	float64	float64	0	0	Glycated haemoglobin (%). Elevated (>7%) for diabetic patients. Range: 4.0–15.0. Key feature for endocrine-related readmissions
creatinine	float64	float64	0	0	Serum creatinine (mg/dL).

					Elevated (>1.5) for CKD patients. Range: 0.4–15.0
haemoglobin	float64	float64	0	0	Blood haemoglobin level (g/dL). Lower values reflect anaemia, BPL status, and comorbidities. Range: 4.0–18.0.
systolic_bp	integer	integer	0	0	Systolic blood pressure (mmHg) at admission. Elevated for cardiovascular diagnoses. Range: 70–220.
readmitted_30d	integer	integer	0	0	PRIMARY TARGET. 1 = patient readmitted within 30 days of discharge. Class rate: ~11.8%. Use class_weight='balanced' or SMOTE.
readmitted_7d	integer	integer	0	0	SECONDARY TARGET. 1 = patient readmitted within 7 days. Subset of readmitted_30d. Harder task (~0.8% positive rate).

Dataset: Diagnoses

Column name	Data Type	Expected Data Type	Missing %	Invalid Value	Description
diag_id	string	object	0	0	Primary key. Unique identifier for each diagnosis record.
admission_id	string	object	0	0	Foreign key → admissions.csv. Links diagnosis to the specific admission.
icd10_code	string	object	0	0	ICD-10-CM diagnosis code (e.g. I21, E11, A41). 34 unique codes spanning 10 clinical categories.
diag_desc	string	object	0	0	Human-readable description of the diagnosis (e.g. 'Acute myocardial infarction', 'Type 2 diabetes mellitus').
diag_rank	integer	int64	0	0	Diagnosis rank within admission: 1 = primary (admission reason), 2–5 = secondary/comorbid. Most

					admissions have 1–3 diagnoses.
diag_category	string	object	0	0	ICD-10 chapter grouping: Cardiovascular / Respiratory / Endocrine / Genitourinary / Gastrointestinal / Infectious / Injury / Neoplasm / Neurological / Obstetric / Perinatal.

Dataset: Billing

Column name	Data Type	Expected Data Type	Missing %	Invalid Value	Description
bill_id	string	object	0	0	Primary key. Unique billing record identifier
admission_id	string	object	0	0	Foreign key → admissions.csv. One bill per admission
total_cost_inr	integer	int64	0	0	Total hospital bill in Indian Rupees. Computed as: $\text{daily_rate} \times \text{ward_multiplier} \times \text{los_days} \times \text{procedure_factor}$. ICU avg: ₹3.8L, General avg: ₹86K.
govt_subsidy_inr	integer	int64	0	0	Government subsidy applied (Ayushman Bharat / PM-JAY / ESI). Zero for uninsured patients. Ayushman cap: ₹30K–₹5L depending on procedure.
out_of_pocket_inr	integer	int64	0	0	Patient-borne cost = $\text{total_cost_inr} - \text{govt_subsidy_inr}$. Key target for healthcare affordability analysis. Avg: ₹48,173
cost_category	string	object	0	0	Dominant billing category: Pharmacy (30%) / Procedure (28%) / Room (25%) / Lab (17%).

Dataset: Hospital

Column name	Data Type	Expected Data Type	Missing %	Invalid Value	Description
hospital_id	string	string	0	0	Primary key. Unique hospital identifier
name	string	string	0	0	Hospital name (synthetic). Format: 'Government General Hospital {State}' / 'District Hospital {State}' / 'Apollo/Manipal (Private) {State}'
state	string	string	0	0	Indian state where the hospital is located. One of 15 states.
tier	string	string	0	0	Hospital tier: tier1 = corporate private (highest cost), tier2 = government general, tier3 = district/rural (lowest cost). Daily rate ratio: 8:2.7:1.
beds	integer	integer	0	0	Total bed capacity. Range: 50–1,200. Correlated with tier.
teaching	boolean	boolean	0	0	Whether the hospital has a teaching/medical college affiliation (True/False). All tier-1 hospitals are teaching hospitals.